

BUCHS-AARAU, Switzerland

WMO: 066330

Lat: 47.3844N

Lon: 8.0794E

Elev: 1276

StdP: 14.03

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 17.1	(c) 21.2	(d) 10.4	(e) 9.8	(f) 19.5	(g) 15.2	(h) 12.4	(i) 23.9	(j) 19.4	(k) 46.3	(l) 16.5	(m) 44.2	(n) 4.0	(o) 50	(p) 0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.5	(c) 88.5	(d) 68.1	(e) 84.8	(f) 67.2	(g) 81.4	(h) 65.9	(i) 69.9	(j) 83.7	(k) 68.4	(l) 81.2	(m) 67.1	(n) 78.5	(o) 4.5	(p) 40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.7	99.4	73.8	64.4	94.8	71.9	63.0	90.4	70.8	34.6	84.0	33.4	81.5	32.2	78.6	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	11.9	9.7	11.9	93.5	5.1	2.9	8.2	95.6	5.3	97.3	2.4	98.9	-1.3	101.0
			11.0	73.0	5.3	1.9	7.2	74.4	4.2	75.5	1.2	76.6	-2.6	78.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	34.4	35.6	43.1	51.0	57.6	65.1	68.0	66.7	59.4	51.7	41.8	35.4
	DBStd	13.67	7.12	7.31	7.01	6.61	6.37	6.53	5.51	5.76	5.62	6.55	6.16	6.52
	HDD50	1960	484	402	228	68	10	1	0	0	3	61	252	451
	HDD65	5480	949	822	678	421	243	79	33	48	183	413	695	916
	CDD50	2292	1	0	14	98	245	453	557	519	284	114	7	0
	CDD65	335	0	0	0	0	12	82	125	102	14	0	0	0
	CDH74	3241	0	0	0	26	168	808	1201	907	129	2	0	0
	CDH80	1037	0	0	0	1	28	258	418	319	13	0	0	0
Wind	WSAvg	3.2	3.9	3.8	4.0	3.5	3.4	3.1	2.9	2.6	2.5	2.5	3.1	3.9
Precipitation	PrecAvg	43.2	2.9	2.8	2.8	3.2	4.3	4.4	4.6	4.4	3.3	3.3	3.5	3.6
	PrecMax	55.4	7.3	7.3	9.7	6.9	8.0	8.1	8.1	8.3	5.9	6.1	7.8	7.7
	PrecMin	29.4	0.6	0.9	1.0	0.3	1.2	1.4	1.4	0.7	1.6	0.4	0.2	0.0
	PrecStd	6.4	1.9	1.6	1.9	1.7	1.7	1.7	1.7	1.8	1.3	1.5	1.9	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	58.6	67.5	77.6	83.9	91.6	93.4	93.5	81.7	71.7	61.5	52.9
		MCWB	48.5	47.6	52.1	57.9	64.8	69.5	69.2	68.8	65.6	60.2	53.2	49.1
	2%	DB	50.7	53.1	63.6	72.8	78.3	86.6	88.6	87.4	78.1	67.5	56.4	49.8
		MCWB	46.3	45.0	50.0	55.5	62.0	67.7	67.6	68.4	64.5	58.8	51.5	46.9
	5%	DB	47.5	49.3	59.7	69.0	74.6	82.7	84.6	83.2	73.7	64.2	53.2	47.4
		MCWB	43.9	43.4	48.5	54.3	59.8	66.6	66.8	67.4	62.4	57.3	49.4	44.6
	10%	DB	44.4	45.9	55.3	64.8	70.4	78.5	81.0	78.9	69.8	60.9	50.4	44.8
		MCWB	41.6	41.2	46.8	52.2	57.9	64.7	65.5	66.0	61.0	56.1	47.4	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	48.9	53.6	59.2	65.8	72.5	71.4	71.5	67.8	62.1	54.4	49.7
		MCDB	52.7	55.8	64.4	73.8	79.0	86.3	87.5	85.0	78.5	68.7	59.6	52.1
	2%	WB	46.8	46.0	51.3	57.0	63.7	69.6	69.3	69.7	65.6	60.1	52.0	47.2
		MCDB	50.0	51.8	60.2	70.3	75.5	82.6	83.5	83.1	74.9	65.4	55.9	49.7
	5%	WB	44.0	43.9	49.5	55.0	61.5	67.6	67.9	68.2	63.7	58.3	49.9	45.0
		MCDB	47.3	49.1	57.2	66.4	71.1	79.6	80.8	80.7	71.6	62.9	53.0	47.2
	10%	WB	41.7	41.5	47.6	53.1	59.4	65.7	66.6	66.6	61.9	56.5	47.8	42.5
		MCDB	44.3	45.5	54.7	63.1	68.1	76.5	78.0	77.0	68.3	60.3	50.0	44.6
Mean Daily Temperature Range	5% DB	MDBR	8.8	12.5	17.2	19.5	18.4	19.5	20.5	19.4	17.8	14.1	10.1	8.0
		MCDBR	12.9	19.9	26.0	28.5	26.6	27.3	28.2	27.2	24.2	18.7	14.9	10.5
		MCWBR	9.1	12.5	14.2	13.5	11.9	10.8	10.3	10.4	11.6	10.5	10.0	8.0
	5% WB	MCDBR	11.9	18.0	22.7	25.5	23.0	24.8	24.9	24.4	21.5	17.1	14.1	9.4
		MCWBR	8.8	11.8	12.9	12.4	10.6	10.3	9.7	9.9	10.5	10.0	9.7	7.6
Clear-Sky Solar Irradiance	taub	0.300	0.327	0.363	0.395	0.407	0.411	0.412	0.404	0.381	0.358	0.324	0.297	
	taud	2.438	2.364	2.311	2.257	2.267	2.299	2.289	2.326	2.374	2.418	2.461	2.446	
	Ebn at Noon	250	265	271	272	272	271	269	266	262	249	238	236	
	Edh at Noon	21	28	35	40	41	40	40	37	32	27	21	19	
All-Sky Solar Radiation	RadAvg	366	632	1018	1377	1587	1846	1780	1550	1171	737	405	299	
	RadStd	44	91	146	202	158	171	161	153	116	75	50	50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.67	N/A	N/A	N/A	N/A	N/A
(47) Regional (117 neighbors)	N/A	N/A	N/A	+1.31	+0.77	+0.61	N/A	N/A	+101	+25

Nomenclature: See separate page